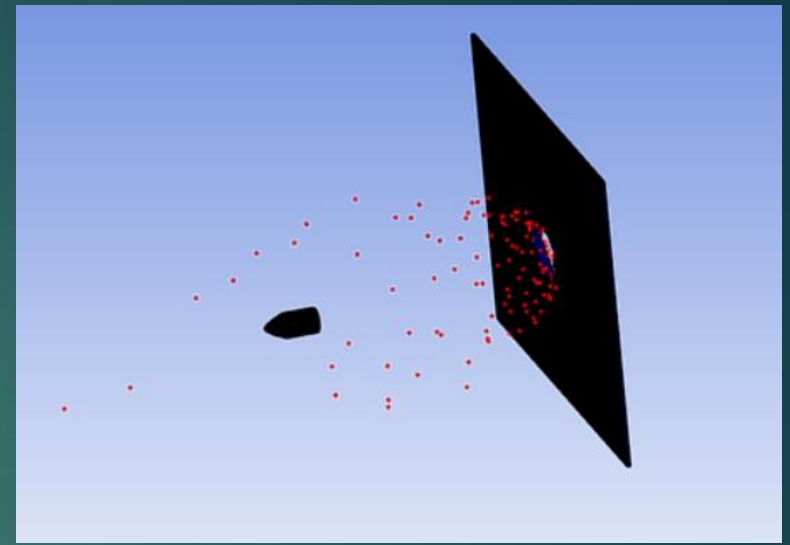
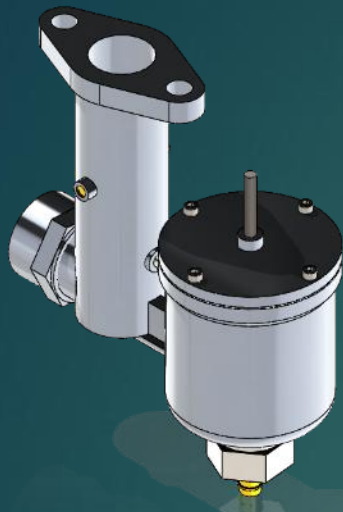




TANAY LAVINGIA'S

PORTFOLIO

ABOUT ME



**FRESINIUS
MEDICAL CARE**

SIEMENS
energy



I am passionate Mechanical Engineer with a strong interest in the field of Automotive and Motorsports engineering, Computer aided design, development and analysis. My passion led me to intern at my university's Formula Student team which contributed to my in-depth understanding of automotive internals. Currently Pursuing Masters in Science in Mechanical Engineering at University of Massachusetts Lowell. With a strong foundation in engineering principles, I am adept at working with cross functional teams to drive innovation and optimize processes. My goal is to continuously learn and advance in the world of engineering, design and development .

EDUCATION

❑ UNIVERSITY OF MASSACHUSETTS LOWELL

Francis College Of Engineering
Massachusetts, USA

Intake
Graduation date

Fall 2024
Dec 2026

CGPA

3.7/4

MS IN MECHANICAL ENGINEERING

Mech 5740: Design For Reliability Engineering

Mech 5760: Engineering Project Management

Mech 5120: Applied Finite Element Analysis

Mech 5200: Numerical Methods for Partial Differential Equations



University of
Massachusetts
Lowell

EDUCATION

❑ GUJARAT TECHNOLOGICAL UNIVERSITY

L.J. Institute of Engineering and Technology
Ahmedabad, India

Intake	Fall 2019
Exp. Graduation date	April 2023
GPA	7.04/10

BE IN MECHANICAL ENGINEERING

Fundamental of Machine Design
Design of Machine Elements
CAD/CAM
Engineering Graphics and Design
Material Science & Metallurgy
Advanced Manufacturing Processes
Quality & Reliability Engineering
Rapid prototyping
Industry 4.0



EXPERIENCE

Fresenius Medical Care North America(Lawrence,Massachusetts)



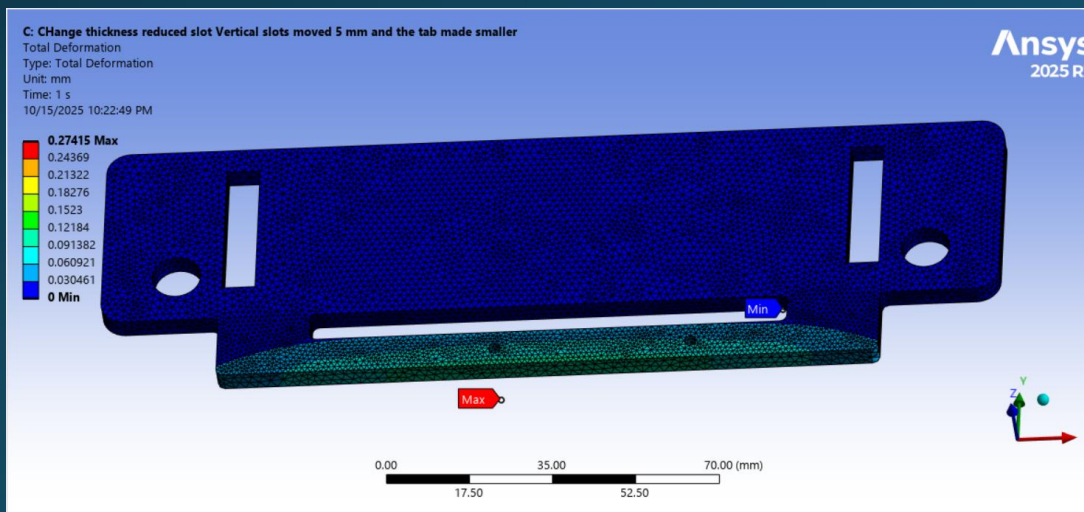
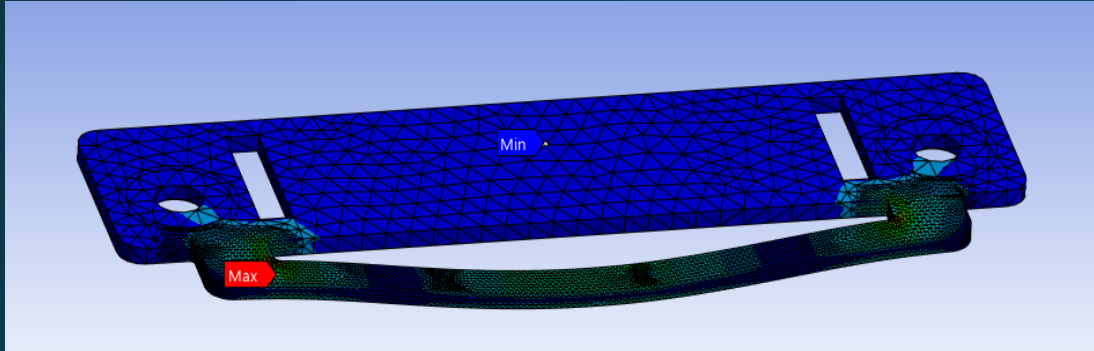
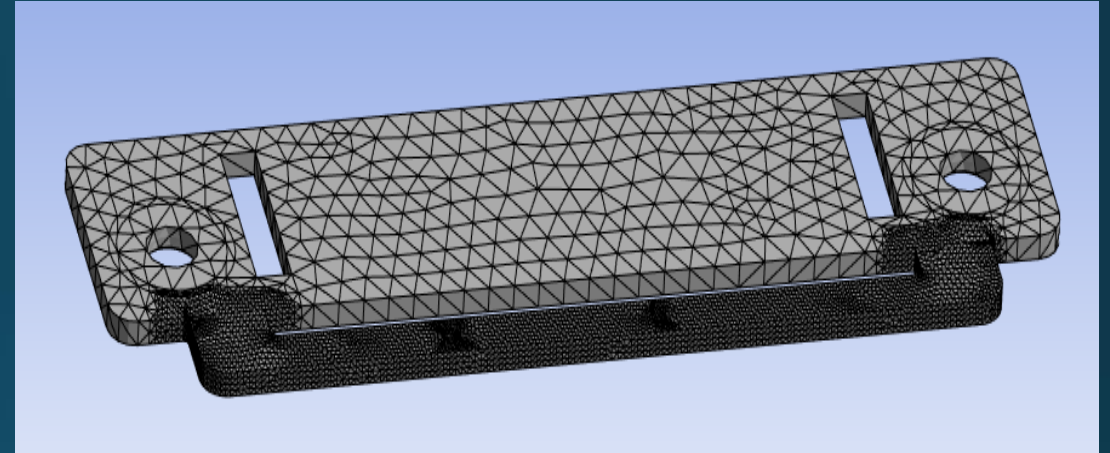
Mechanical Engineering Co-op: July 2025- December 2025

EXPERIENCE



**FRESENIUS
MEDICAL CARE**

- **Problem-** Optimize the design of a platform to ensure that the max stress doesn't cross stress



Performed simulations in Ansys Static Structural on different design iterations to make sure stress generated is under yield stress

EXPERIENCE



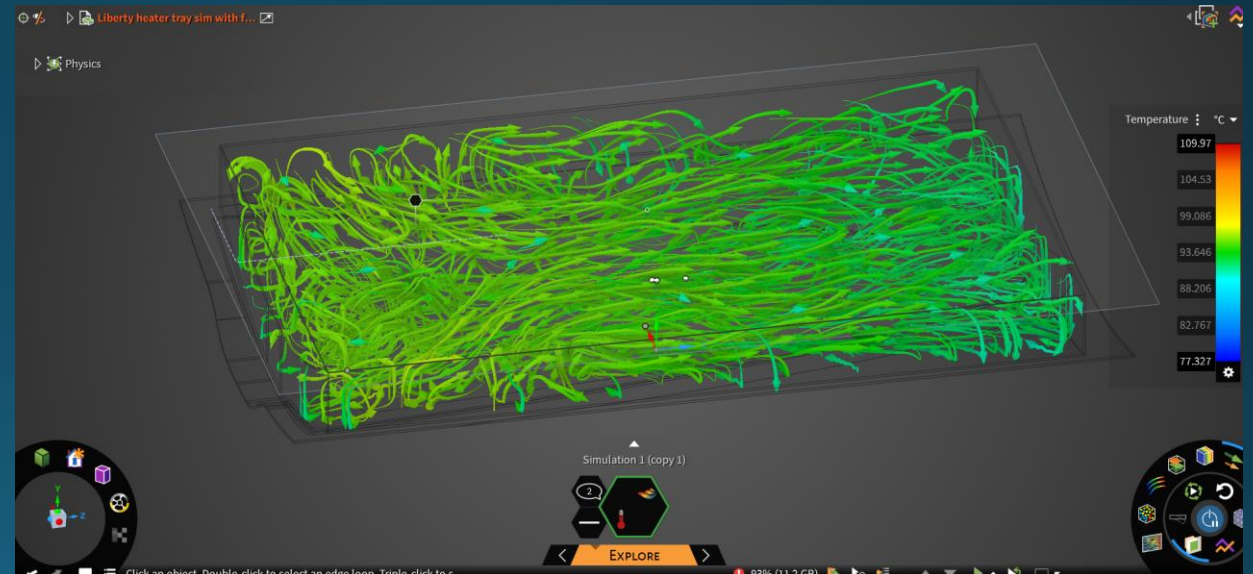
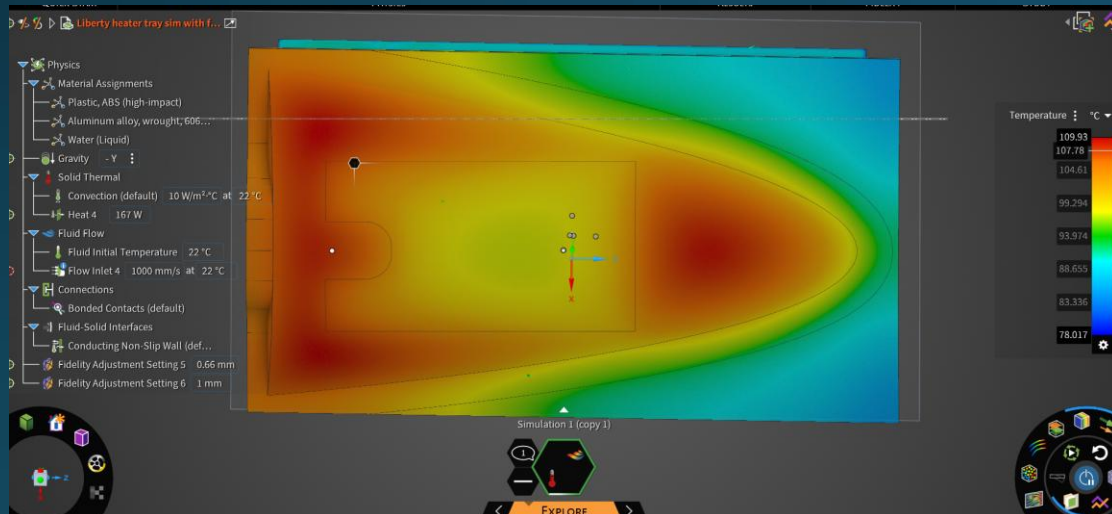
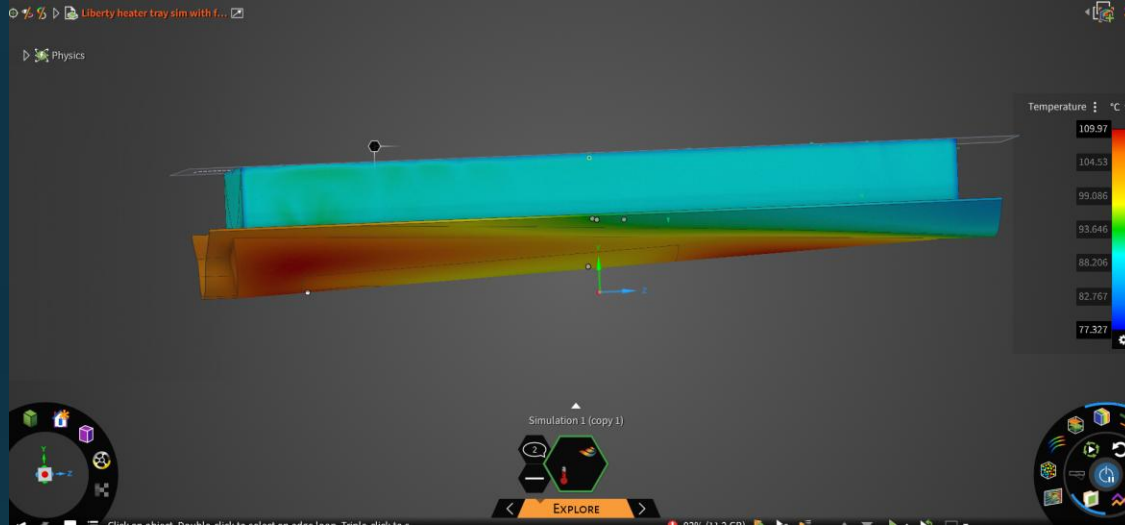
Manufactured one of the designs on a Bridgeport 2-axis milling machine to perform physical testing of the assembly

EXPERIENCE



**FRESENIUS
MEDICAL CARE**

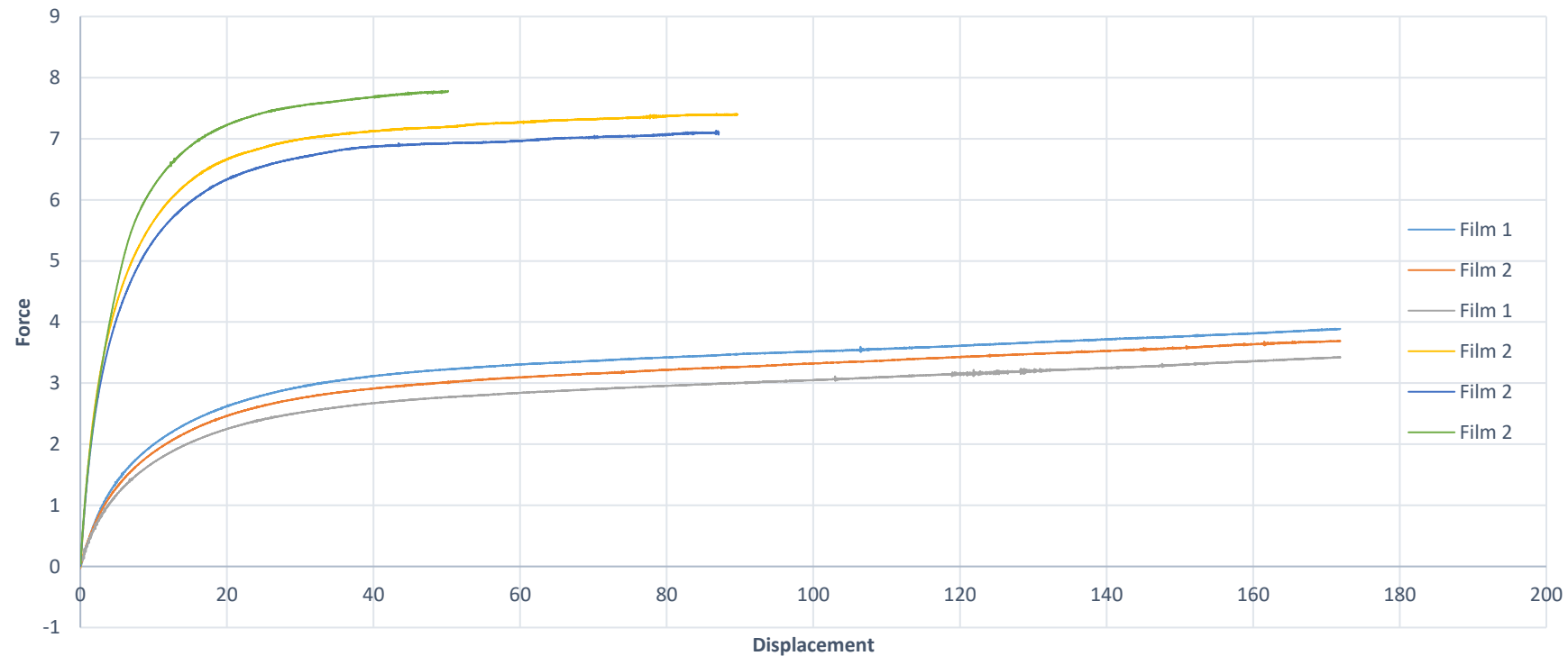
Problem- The heater tray for old Peritoneal Dialysis(PD) Cycler had slow heating, we had to make improvements and make a heater tray for the new PD Cycler.



Performed **Thermofluid** simulation in Ansys on old heater tray Assembly to get data on the heating time, find out the scope of Improvement and optimize design for new cycler.

EXPERIENCE

Performed tensile testing on 2 different plastic films to obtain material properties and use it in a simulation to compare the performance.

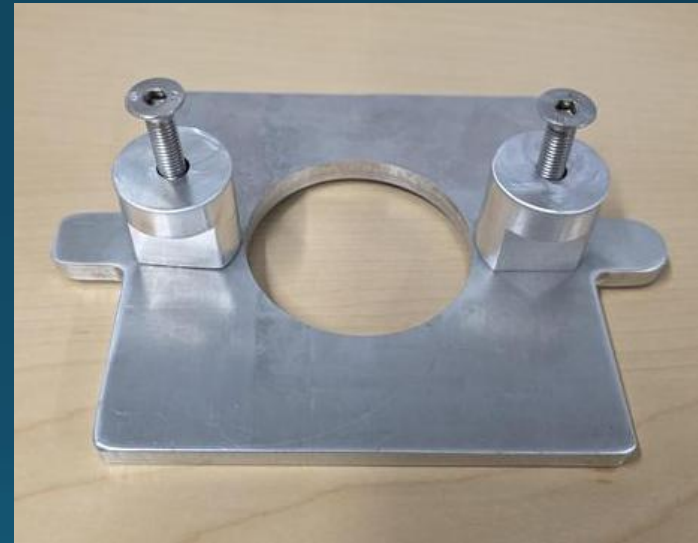
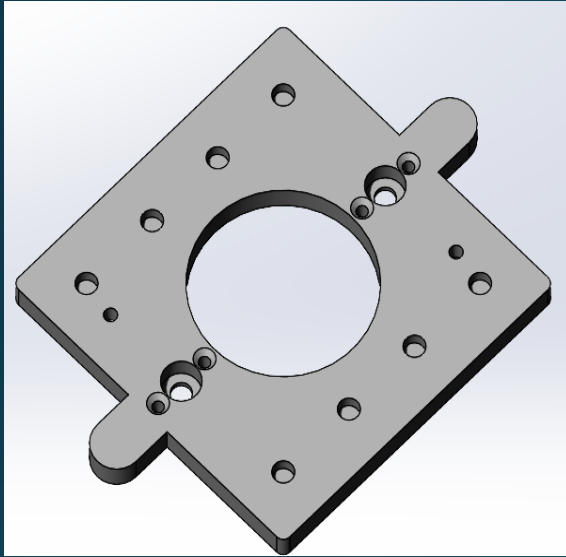


Tensile testing on plastic films

EXPERIENCE



**FRESENIUS
MEDICAL CARE**

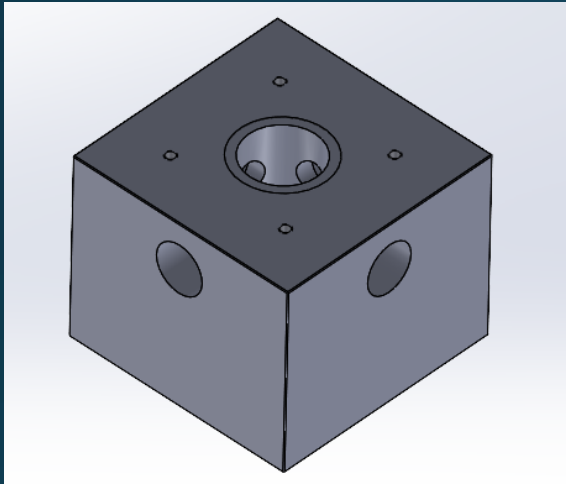


Clamping plate for plastic film to clamp
the plate to a fixture
Designed in SolidWorks and machined
out of Aluminum on Bridgeport 2-axis
mill

EXPERIENCE



**FRESENIUS
MEDICAL CARE**

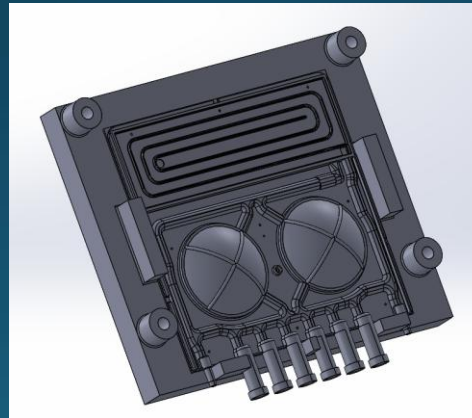
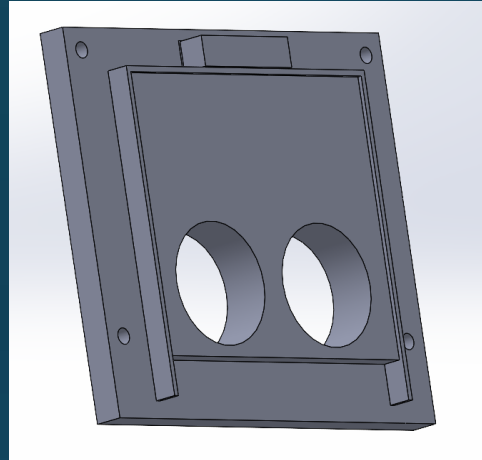
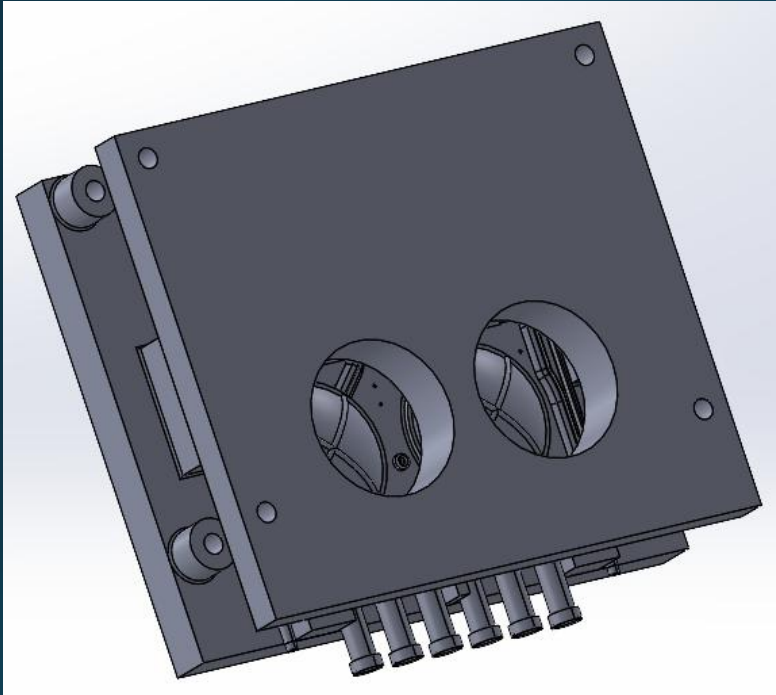


Fixture designed to mount a
pressure sensor with four outlets
Designed in SolidWorks and
Machined out of Aluminum on
Bridgeport 2-axis mill

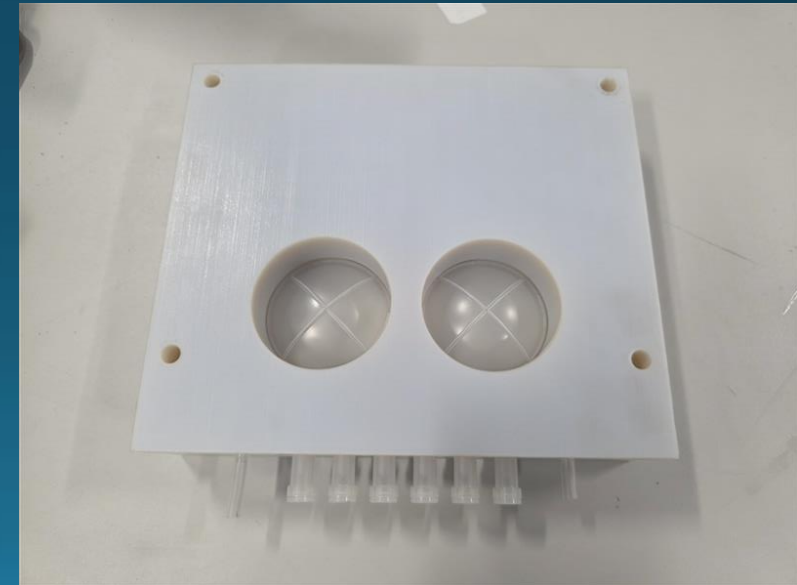
EXPERIENCE



**FRESENIUS
MEDICAL CARE**



Fixture for clamping the
disposable(called Cassette) of
Peritoneal Dialysis Cyclor
Designed on SolidWorks and 3d
Printed on an Objet printer



EXPERIENCE

3Brothers Machination PVT LTD(Ahmedabad, India)

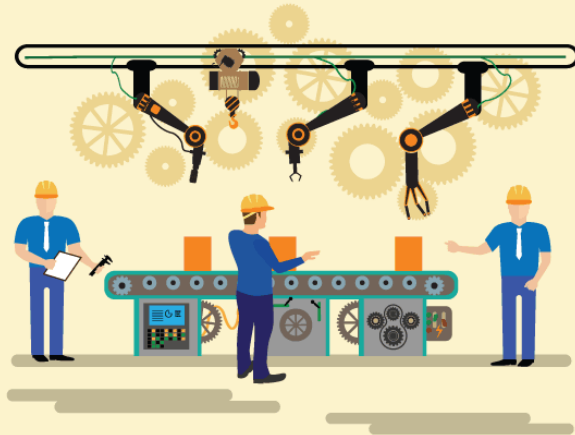


Mechanical Engineering Intern – January 2023-June 2023

Project Coordinator- Machine Design and Development – July 2023- July 2024

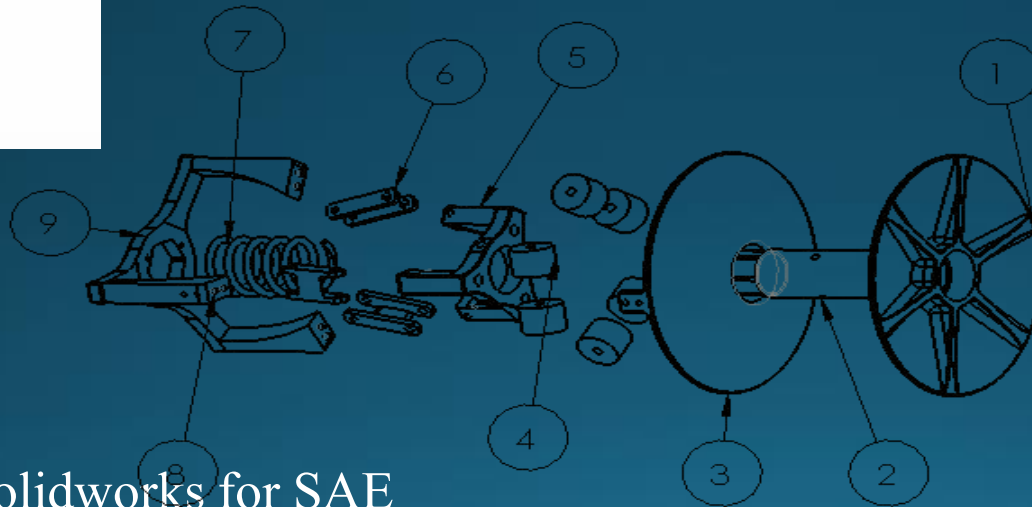
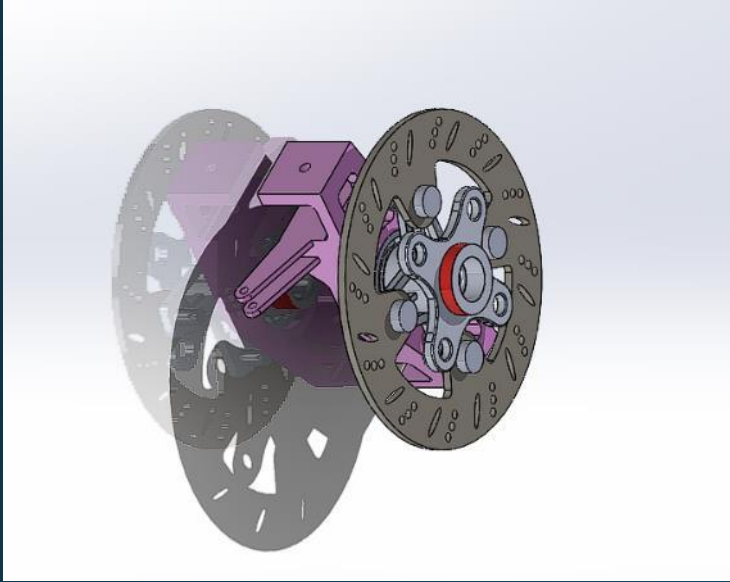
EXPERIENCE

Led an interdisciplinary team to develop a Hydraulic Cold Forging Machine for TMT rod to increase its efficiency and reliability.

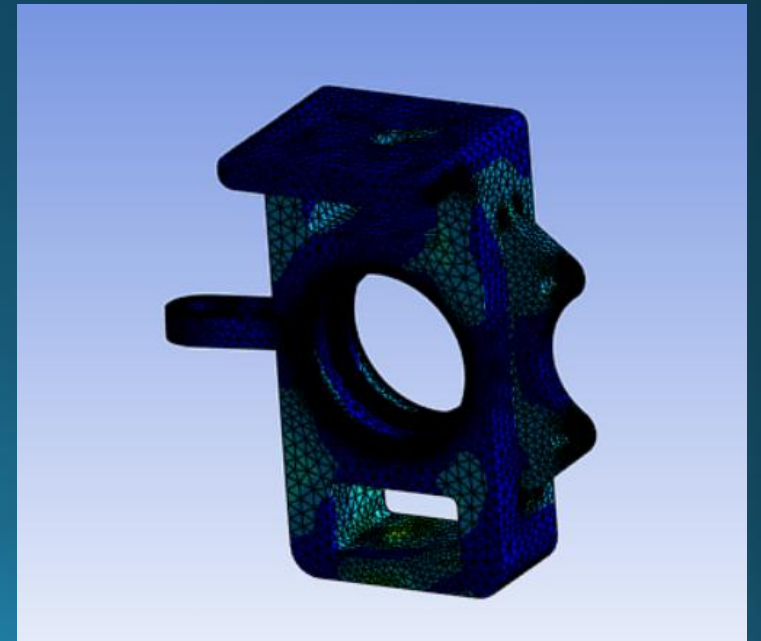


Optimized and streamlined manufacturing processes for reduction in lead time.

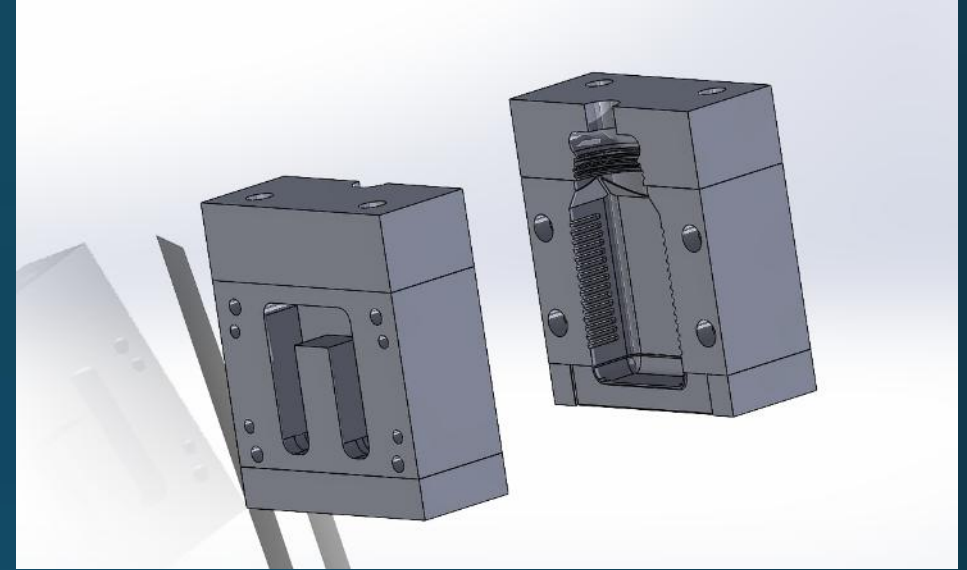
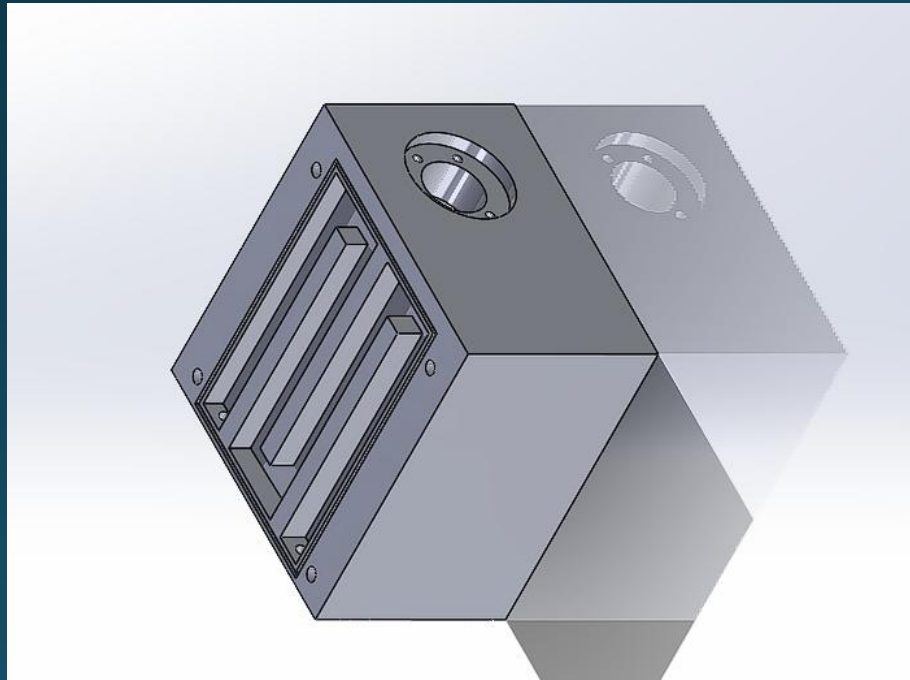
EXPERIENCE



Developed 3D models in Solidworks for SAE vehicles, made their 2D drawings and did Structural Analysis in ANSYS



EXPERIENCE



Designed blow molding and sheet forming dies on SolidWorks and supported production by designing fixtures for them.

INTERNSHIP

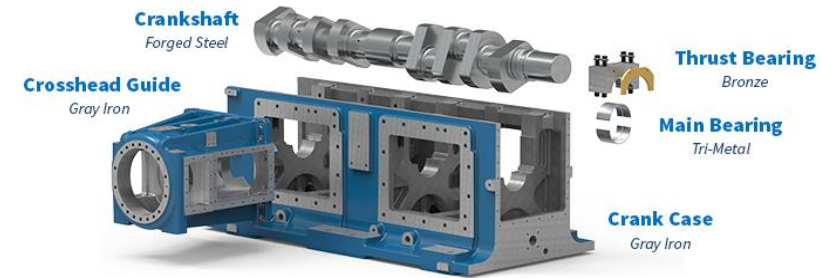
SIEMENS
energy

Student Intern

June 2022-July 2022

INTERNSHIP

- Gained Knowledge on design, mechanisms and auxiliary systems of Reciprocating Compressors.
- Observed various NDTs (Non-Destructive Tests) used to inspect high pressure compressor components and analyzed test result.
- Worked with production team and updated operation manager on daily production activities.



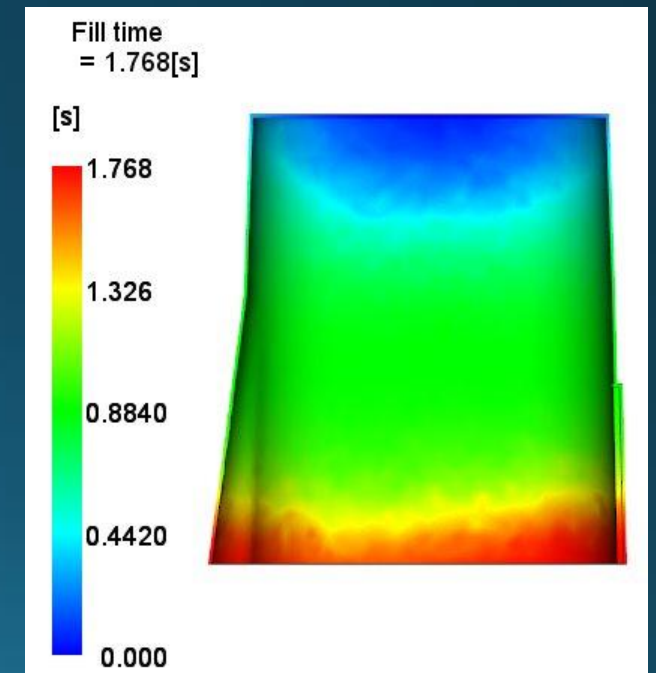
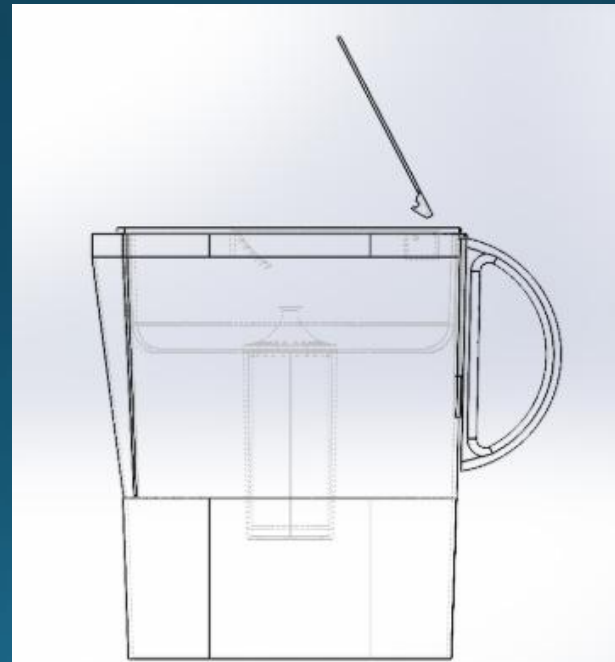
INTERNSHIP



PROJECTS

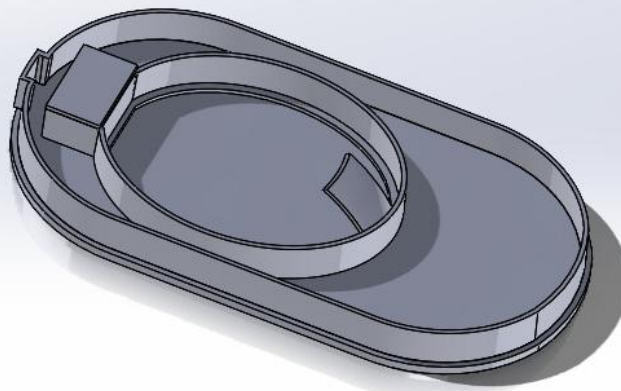
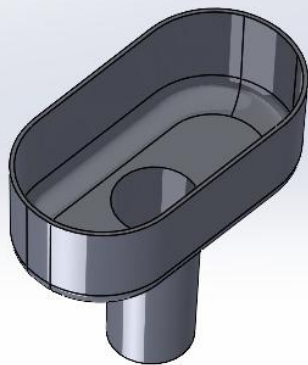
Course: Plastic Product Design

Case Study of a water filtration system



PROJECTS

- CAD of individual components and assembly



PROJECTS

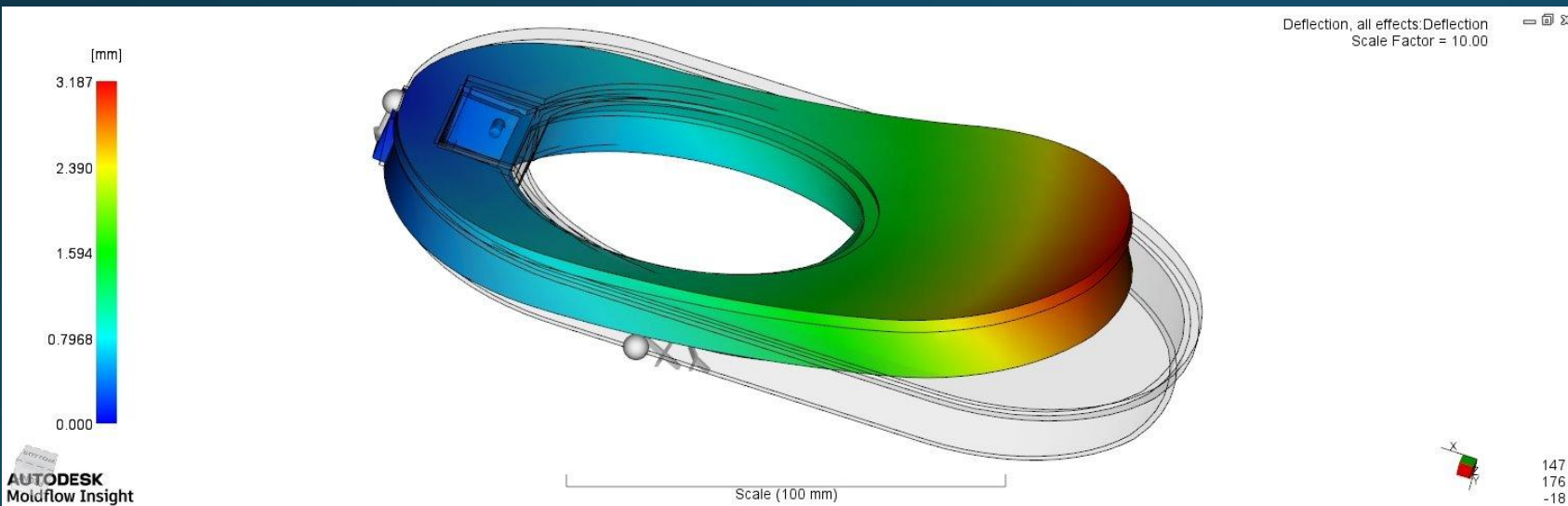
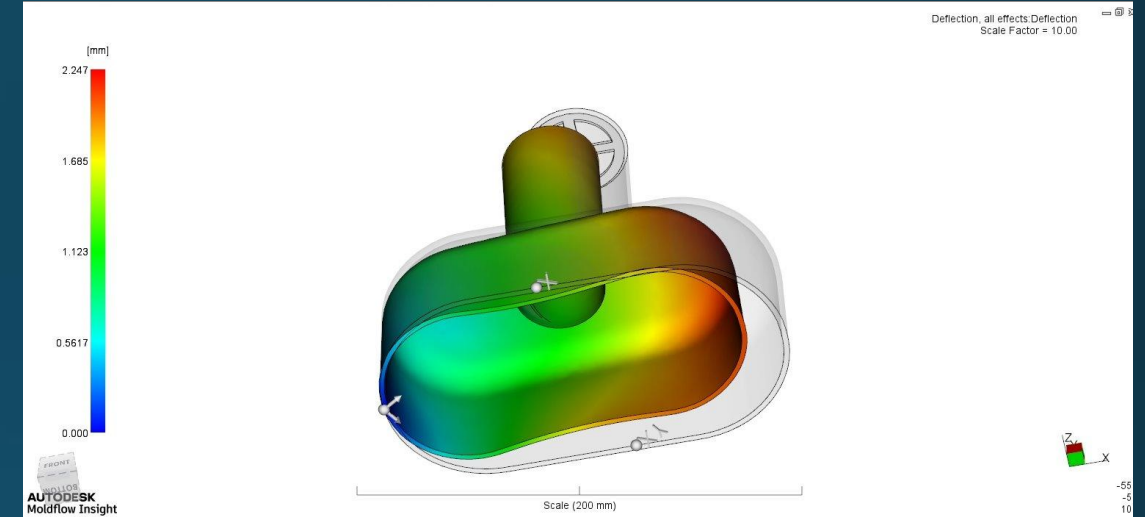
GROUP A — Colored Handles & Lids (pigmentation, stiffness, snap-fit)								
Material / Trade Name	Plastic Type	Density (g/cm³)	Tensile Str. (MPa)	Flex. Mod. (MPa)	Notched Izod (J/m)	Heat Defl. °C @1.8 MPa	Food Contact Status	Est. Price (\$/lb)
BASF Ultraform N2320 003 PRO AT	Acetal / POM copolymer	1.41	64	2700	~55	95	USP VI; ISO 10993-5; FDA DMF	\$1.48/lb
Americas Styrenics STYRON 678D	Polystyrene – high heat	1.04	37.2	3380	~21	80	Food contact compliant	\$0.72/lb
LyondellBasell Pro-fax PD702	PP – homopolymer (high flow)	0.9	31	1170	32	88 (ASTM D648 @0.45 MPa)	FDA 21 CFR 177.1520 (incl. cooking); EU 10/2011	\$0.70/lb

Material Selection according to the requirements

GROUP B — Clear Basin (optical clarity, food/water contact)								
Material / Trade Name	Plastic Type	Density (g/cm³)	Tensile Str. (MPa)	Flex. Mod. (MPa)	Notched Izod (J/m)	Heat Defl. °C @1.8 MPa	Food Contact Status	Est. Price (\$/lb)
SABIC LEXAN HF1140 (Europe)	Polycarbonate (PC)	1.2	50 / 63	2300	640	121	FDA food contact (limited colors)	\$1.10/lb
Eastman Tritan TX1001	Copolyester (amorphous)	1.18	43 / 58	1550	980	85	FDA; NSF/ANSI 51 & 61	\$2.20/lb
Reliance HD50MA180	HDPE	0.95	22	900	30	— (Vicat 123°C)	IS:10146; FDA §177.1520	\$0.58/lb

PROJECTS

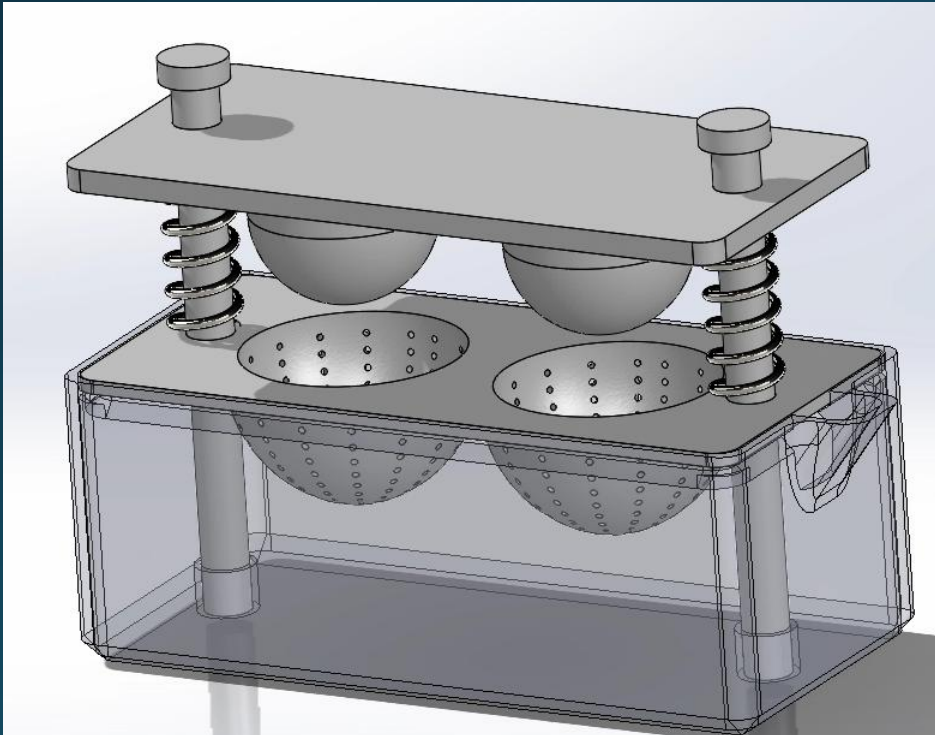
- Moldflow simulations to analyze various behavior and defects



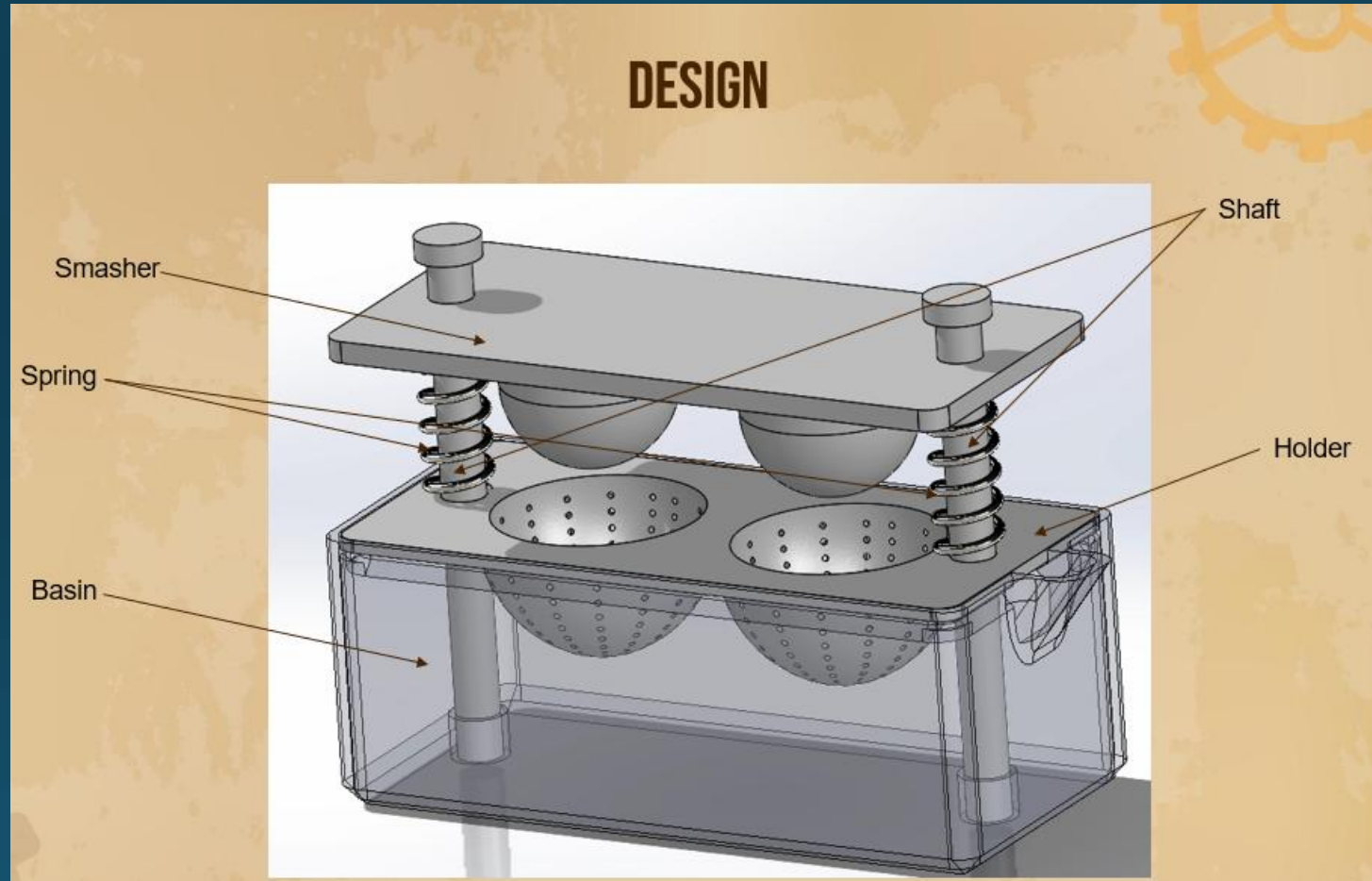
PROJECTS

Course: Rapid Prototyping

Design and Prototyping of a dual lemon juicer



PROJECTS



Designed a multi component lemon/lime juicer

PROJECTS



Prototyping after tolerance analysis

PROJECTS



Failure in Initial Prototype



Implemented design changes



PROJECTS

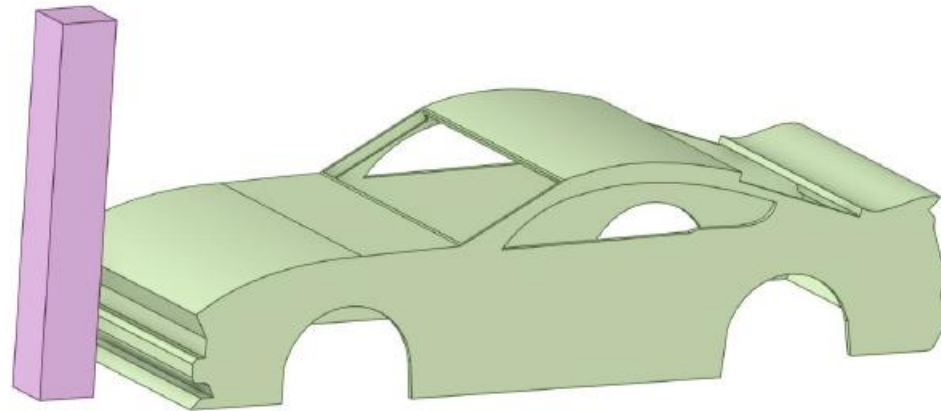


Final Prototype

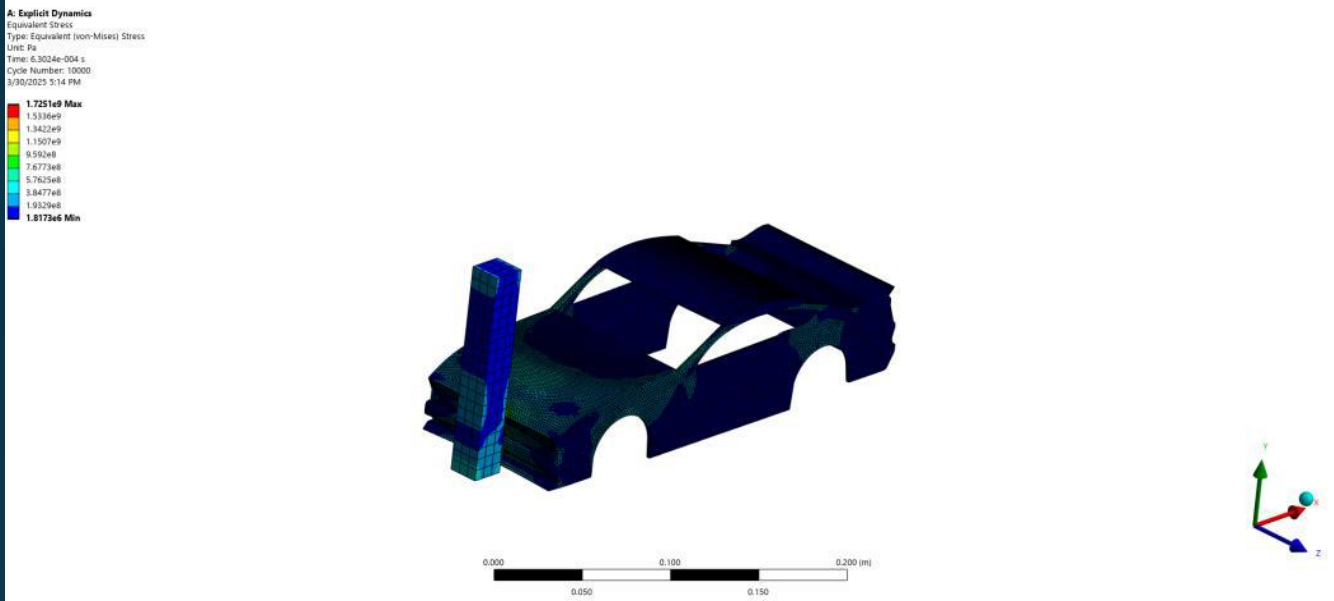
PROJECTS

Course: Applied Finite Element Analysis

Crash test on a car body shell using **Explicit Dynamics** in Ansys

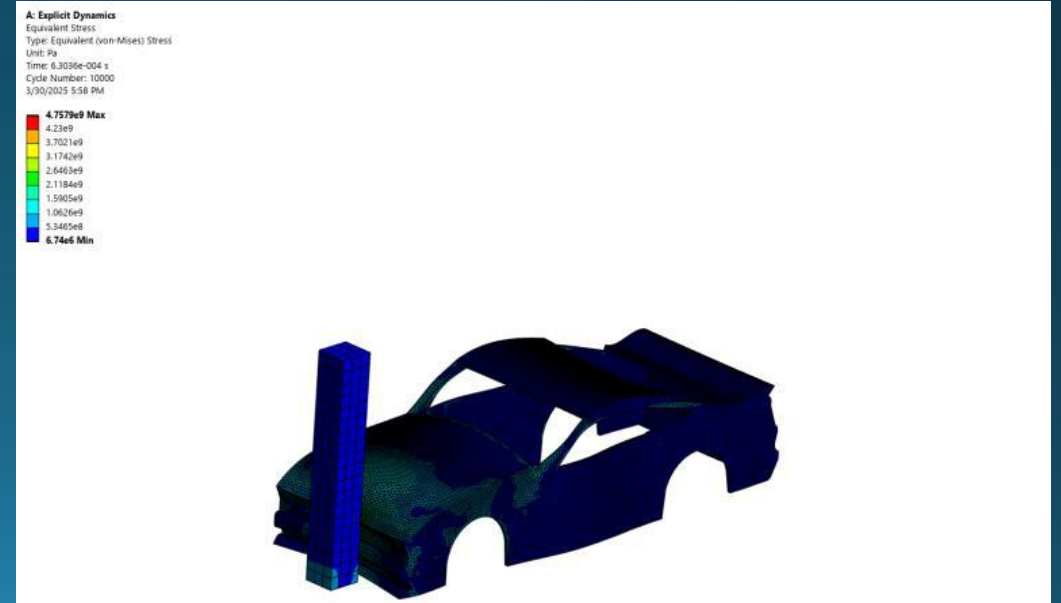


PROJECTS



At 27.8 m/s

At 80 m/s



PROJECTS

Course: Design for Reliability Engineering

FRACAS Analysis on an Elevator

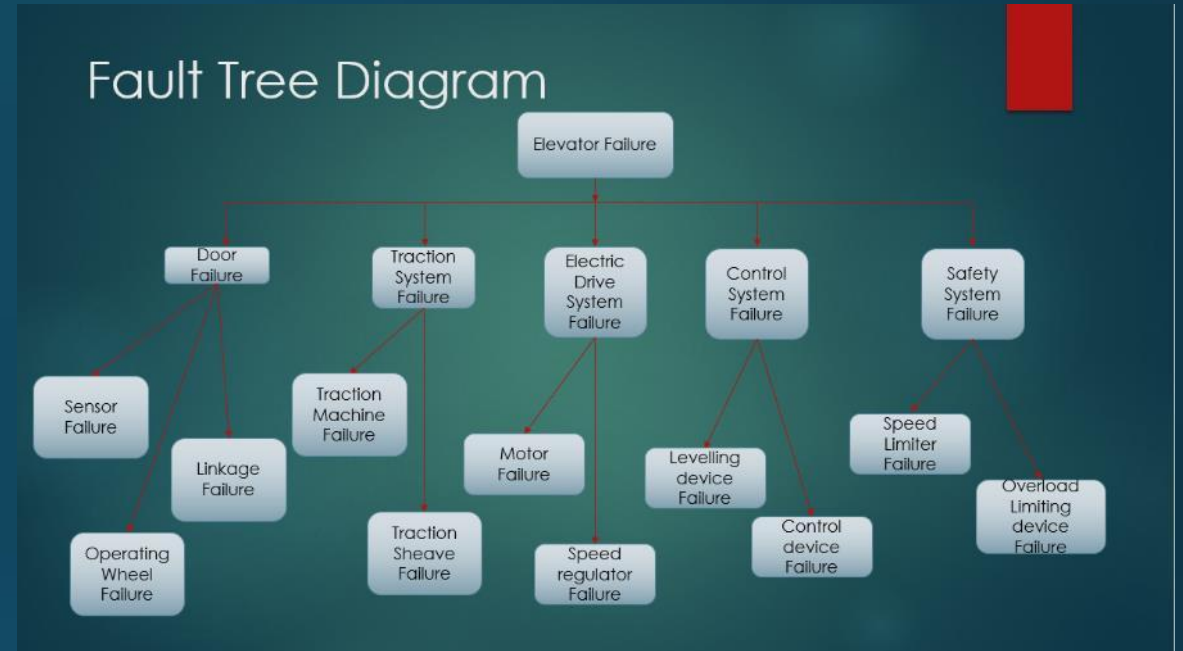
By-Tanay Ketan
Lavingia

UML ID- 02193514



PROJECTS

- Identified types of failure in an elevator and constructed a Fault Tree Diagram based on them.
- Performed calculations to obtain MTBF, MTTR and Availability.
- Performed Failure Modes and Effective Analysis (FMEA) with corrective actions to improve the reliability of the system.

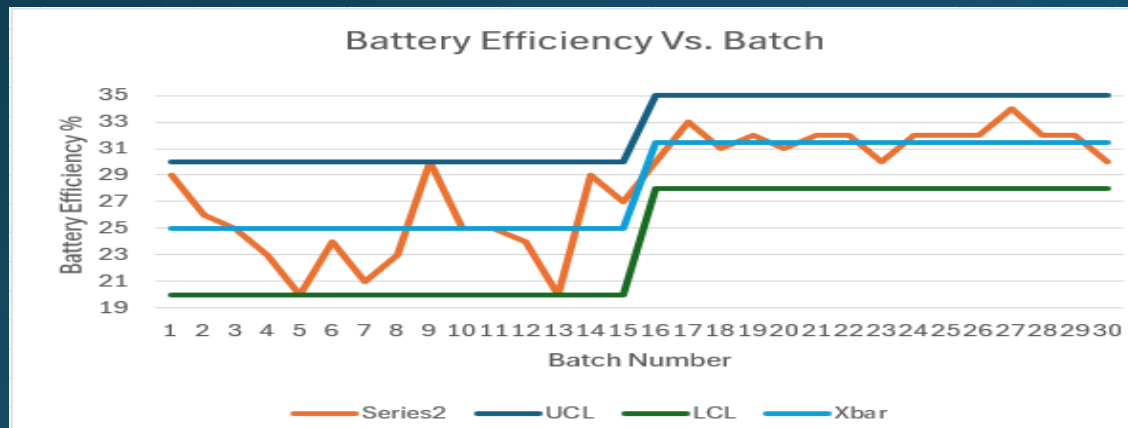


Failure	Mode	Effect	Analysis
F 	M 	E 	A 
Potential	Types, ways, possibilities	Negative Effect on process under study	Study risk and reduce it

PROJECTS

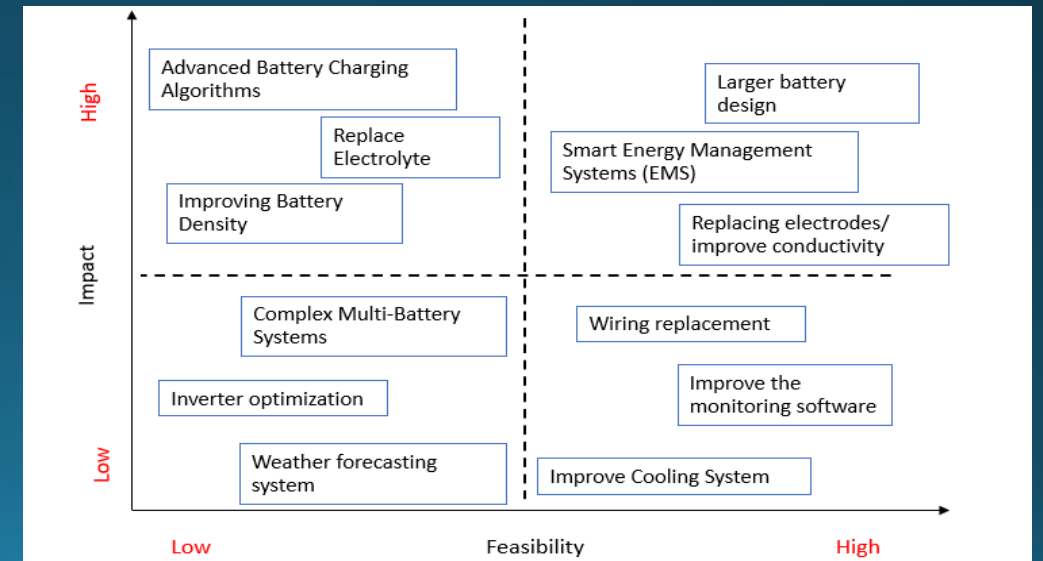
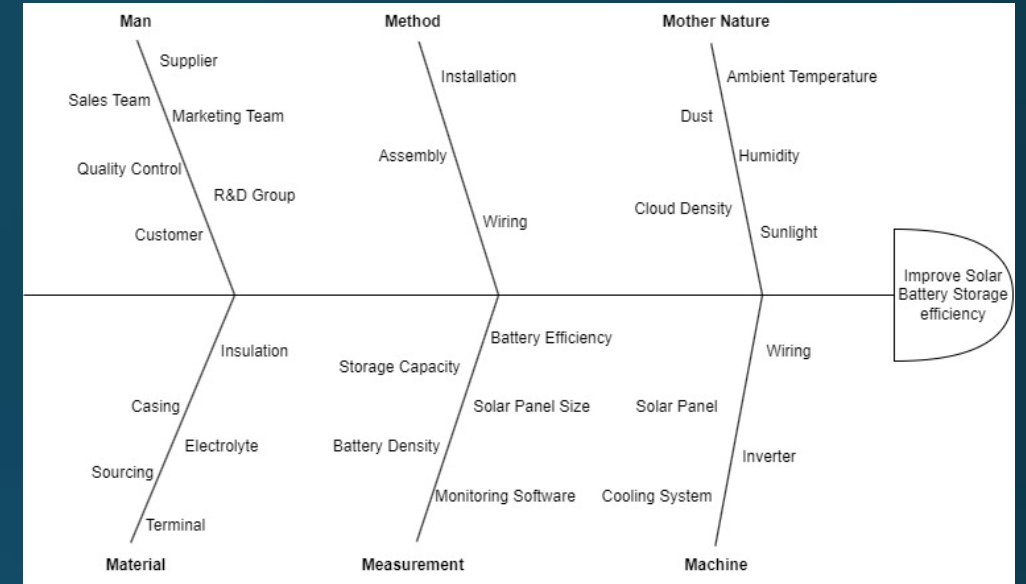
Course: Engineering Project Management

Battery Efficiency Enhancement for Solar Energy Storage



• PROJECTS

- Used DMAIC (Define, Measure, Analyze, Improve, Control) to enhance the storage efficiency for battery used for storage of solar energy from 70 % to 85%.
- Used Ishikawa Diagram to identify the different causes for the reason behind low battery efficiency.
- Used Priority Matrix to narrow down the major causes and used Minitab to do Design of Experiments on them.
- Decided on what aspects of battery need change to improve the efficiency.
- Made control chart in Minitab to check whether the process is in control or not.



SOFTWARES



Tanay Lavingia



Address

Lowell, Massachusetts



Contact

(978)-569-8959



Email

tklavingia@gmail.com